DATA SCIENCE CAPSTONE PROJECT REPORT

Book Pricing & Rating Analysis

PROJECT OBJECTIVE

Objective

The objective of this project was to collect real-world book data from the web, clean and preprocess the raw data, perform exploratory data analysis (EDA), build a machine learning model, and create an interactive Power BI dashboard for business insights.

The project demonstrates how unstructured web data can be transformed into decision-ready business intelligence.

This project focuses on analysing:

- Book pricing patterns
- Rating behaviour
- Category performance
- Price band impact on ratings
- Prediction of high-rated books using Machine Learning

The final goal was to answer:

Can book characteristics such as price, category, and title length help predict whether a book will receive a good rating?

PHASE 1: DATA COLLECTION METHOD (WEB SCRAPING)

Data Source

Website Used:

Books to Scrape
http://books.toscrape.com/

This website was selected because:

- It is publicly accessible
- It allows web scraping
- It contains structured product data

- It provides multiple useful features such as title, price, rating, category, and stock availability

It is ideal for an end-to-end data science capstone project.

Data Extraction Method

Python libraries used:

- Requests
- BeautifulSoup
- Pandas

The scraping process involved:

- Visiting multiple pages of the website
- Extracting book details from each page
- Visiting individual product pages for category extraction
- Collecting more than 1000 records

Extracted columns:

- Title
- Price
- Rating
- Availability
- Product URL
- Category

Final scraped records:

Total Books Collected = 1000

This satisfied the project requirement of collecting at least 1000 clean records.

PHASE 2: DATA CLEANING & PREPROCESSING

Cleaning Steps Performed

The raw scraped data was cleaned and standardized before analysis.

1. Missing Value Check

Checked all columns for null values.

Result:

No missing values found

2. Duplicate Removal

Duplicate records were checked and removed.

Result:

No duplicate records found

Shape remained:

Before: (1000, 6)
After: (1000, 6)

3. Data Type Standardization

Fixed data types for proper analysis:

- Price → converted to float
- Rating → converted to integer

This improved analytical accuracy.

4. Availability Column Removal

The Availability column contained only:

“In Stock”

Since all books had the same value, it provided no analytical value and was removed.

Feature Engineering

Created useful business features:

Price_Band

Books were classified into:

- Low

- Medium
- High

to analyze pricing patterns more effectively.

Rating_Category

Ratings were grouped into:

- Good
- Average
- Poor

to improve business storytelling and ML classification.

Rating_Score

Converted ratings into scaled score format for ML support.

Example:

5-star → 1.0

Title_Length

Calculated number of characters in each book title for predictive analysis.

Final dataset after preprocessing:

1000 rows × 11 columns

The dataset became clean, structured, and ready for EDA and ML modeling.

PHASE 3: EXPLORATORY DATA ANALYSIS (EDA)

Summary Statistics

Price

- Average Price = **35.07**
- Median Price = **35.98**
- Minimum Price = **10.00**
- Maximum Price = **59.99**

This shows that book prices are fairly evenly distributed across low to high price ranges.

Rating

- Average Rating = **2.92**
- Median Rating = **3**
- Most ratings fall between **2 and 4**

This indicates that most books receive average ratings rather than excellent ratings.

Rating Distribution

Distribution of ratings:

- Rating 1 → 226 books
- Rating 2 → 196 books
- Rating 3 → 203 books
- Rating 4 → 179 books
- Rating 5 → 196 books

Observation:

Low-rated books slightly outnumber highly rated books, showing that excellent ratings are less common.

Price Band Distribution

Books by price band:

- High → 403 books
- Medium → 401 books
- Low → 196 books

Observation:

Most books fall into Medium and High price bands, while Low-price books are comparatively fewer.

Category-wise Rating Analysis

Top categories by average rating:

- Adult Fiction → 5.0
- Erotica → 5.0
- Novels → 5.0
- Christian Fiction → 4.2

Lowest categories:

- Crime → 1.0
- Psychology → 1.7
- Parenting → 2.0

Observation:

Book performance differs significantly across categories.

Category plays a stronger role than price in influencing ratings.

Price vs Rating Relationship

Scatter plot and boxplot analysis showed:

- No strong correlation between price and rating
- Expensive books do not necessarily receive better ratings

Correlation between Price and Rating:

Only 0.028

This proves price has almost no predictive relationship with ratings.

Correlation Heatmap

Strong findings:

- Price vs Rating → Very weak correlation
- Title Length vs Rating → Very weak correlation
- Rating vs Rating Score → Perfect correlation (expected)

This confirmed that strong predictive features were limited.

PHASE 4: MACHINE LEARNING MODEL

Problem Statement

The goal was to predict:

Whether a book is "Good Rated" or not

Target variable:

- Good Rated → 1
- Not Good Rated → 0

This was treated as a:

Classification Problem

Why Logistic Regression Was Chosen

Logistic Regression was selected because:

- The target variable is binary (0 or 1)
- It is simple, interpretable, and suitable for classification
- It performs well for baseline predictive modeling
- It helps explain feature influence clearly

Since the project focuses on business understanding and model interpretation, Logistic Regression was the most appropriate starting model.

Features Used

Input features:

- Price
- Title_Length
- Category_Code

These were selected because they were the most relevant structured predictors available from the dataset.

Train-Test Split

Dataset split:

- Training Data → 80%
- Testing Data → 20%

This ensured proper model validation.

Initial Model Problem

Initially, the model predicted only the majority class and ignored the minority class.

Accuracy looked acceptable but prediction quality was poor.

This happened because the dataset was imbalanced.

Improvement Applied

Used:

class_weight = "balanced"

This reduced majority class bias and improved prediction across both classes.

This created a fairer and more realistic model.

Final Model Performance

Accuracy

51.5%

Confusion Matrix

Actual / Predicted	Not Good	Good
Not Good	61	59
Good	38	42

Classification Performance

Good Class

- Precision = 42%
- Recall = 53%

Observation:

The model predicts both classes after balancing, but performance remains moderate.

Why Accuracy Was Only 51.5%

The model accuracy remained moderate because:

1. Weak Feature Relationships

EDA showed:

- Price has almost no correlation with rating

- Title Length has almost no correlation with rating

These features are weak predictors.

2. Limited Predictive Variables

Important features like:

- Reviews
- Author popularity
- Reader engagement
- Publication trends

were not available in the dataset.

These could strongly influence ratings.

3. Category Helps, But Not Enough

Category contributed more than Price and Title Length, but alone it was not strong enough for high predictive accuracy.

Final Interpretation

The model successfully proved an important business insight:

Book ratings cannot be accurately predicted using only price, title length, and category.

This is a strong analytical finding and a valid project conclusion.

PHASE 5: DASHBOARD SUMMARY (Power BI)

An interactive Power BI dashboard was developed to present the complete business insights and machine learning outcomes of the Book Pricing & Rating Analysis project. The dashboard was divided into 3 pages to ensure clear storytelling from summary insights to category-level analysis and final model performance.

The dashboard helps answer:

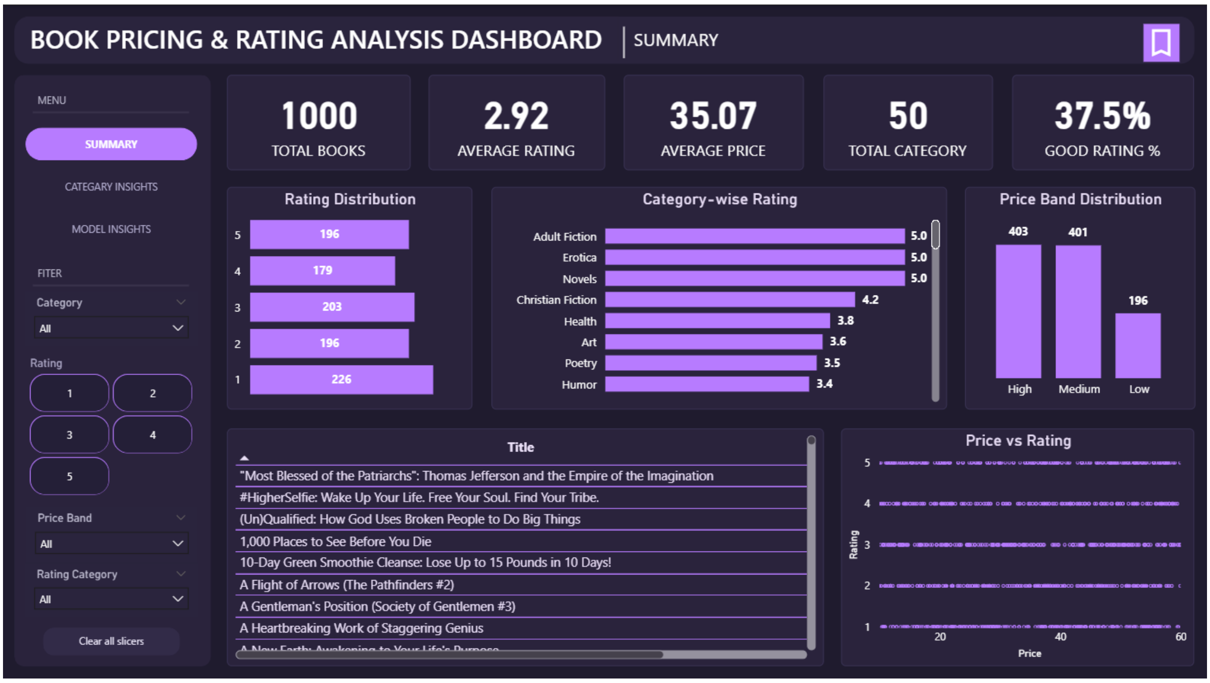
- What is happening in the dataset?
- Which categories perform best and worst?
- Can book ratings be predicted using Machine Learning?

The dashboard includes KPI cards, trend analysis, filters, bookmark functionality, and model performance reporting.

PAGE 1: SUMMARY DASHBOARD

Objective

The Summary Dashboard provides an overall view of the complete dataset and highlights the key business metrics related to pricing, ratings, and category performance.



Key KPIs

Total Books

1000

A total of 1000 books were scraped and analyzed from the Books to Scrape website

Average Rating

2.92

The average book rating is 2.92 out of 5, indicating that most books are rated in the average range rather than highly rated.

Average Price

35.07

The average price of books is 35.07, showing that most books fall within medium to high price ranges.

Total Categories

50

The dataset contains 50 unique book categories, providing strong diversity for analysis.

Good Rating %

37.5%

Only 37.5% of books were classified as Good Rated, showing that highly rated books are comparatively limited.

Key Visual Insights

Rating Distribution

Books with Rating 1 were highest at 226 books, while Rating 5 books were 196.

This shows that low-rated books slightly outnumber highly rated books.

Price Band Distribution

- High Price Band → 403 books
- Medium Price Band → 401 books
- Low Price Band → 196 books

This indicates that most books are concentrated in Medium and High price segments.

Category-wise Rating

Top-performing categories include:

- Adult Fiction → 5.0
- Erotica → 5.0
- Novels → 5.0
- Christian Fiction → 4.2

This proves that category influences rating more than pricing.

Price vs Rating

The scatter plot confirmed no strong relationship between price and rating.

Expensive books do not necessarily receive better ratings.

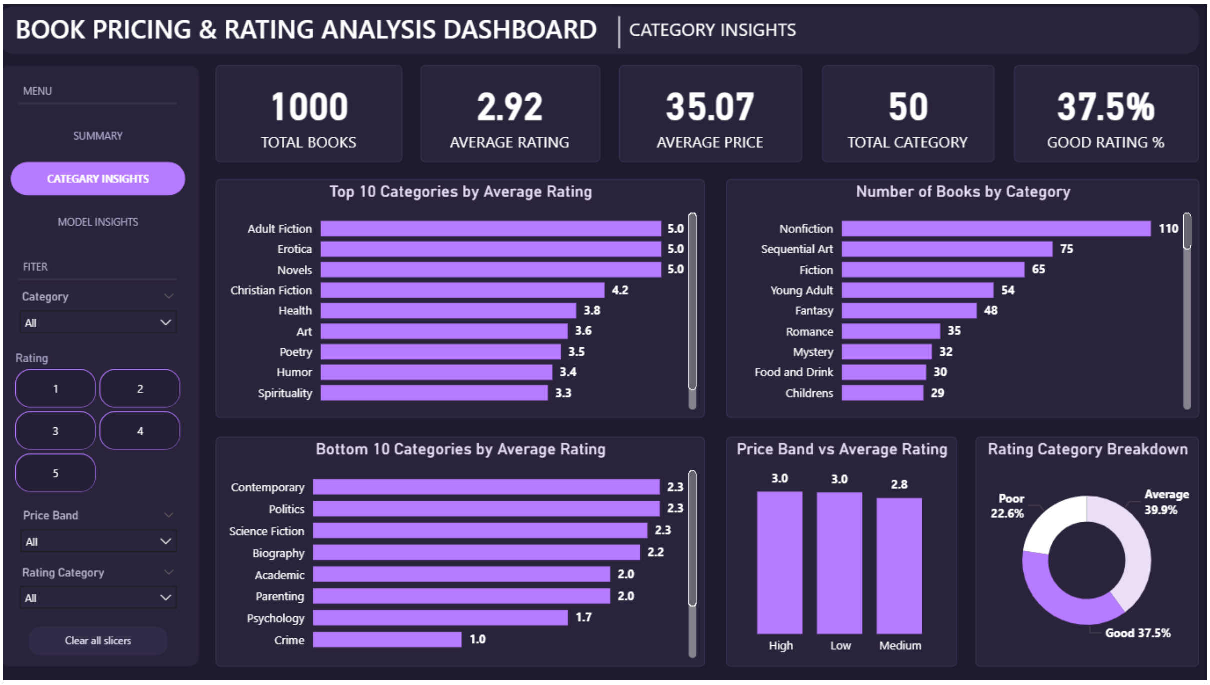
Conclusion

Page 1 provides a high-level understanding of the business problem and establishes that category is a stronger driver of ratings than price.

PAGE 2: CATEGORY INSIGHTS DASHBOARD

Objective

The Category Insights Dashboard focuses on understanding category-level performance, identifying top-performing and low-performing categories, and comparing popularity with rating quality.



Top 10 Categories by Average Rating

Highest-rated categories include:

- Adult Fiction → 5.0
- Erotica → 5.0
- Novels → 5.0
- Christian Fiction → 4.2
- Health → 3.8

These categories show the strongest customer satisfaction and rating consistency.

Bottom 10 Categories by Average Rating

Lowest-rated categories include:

- Crime → 1.0
- Psychology → 1.7
- Parenting → 2.0

- Academic → 2.0
- Biography → 2.2

These categories require quality improvement strategies.

Number of Books by Category

Most common categories include:

- Nonfiction → 110 books
- Sequential Art → 75 books
- Fiction → 65 books
- Young Adult → 54 books

This helps compare popularity versus quality.

For example, some highly rated categories are not the most populated categories.

Price Band vs Average Rating

- High → 3.0
- Low → 3.0
- Medium → 2.8

This confirms that price bands have very little effect on average ratings.

Higher pricing does not guarantee better customer satisfaction.

Rating Category Breakdown

- Average → 39.9%
- Good → 37.5%
- Poor → 22.6%

Most books fall under Average and Good ratings, while Poor-rated books are comparatively fewer.

Conclusion

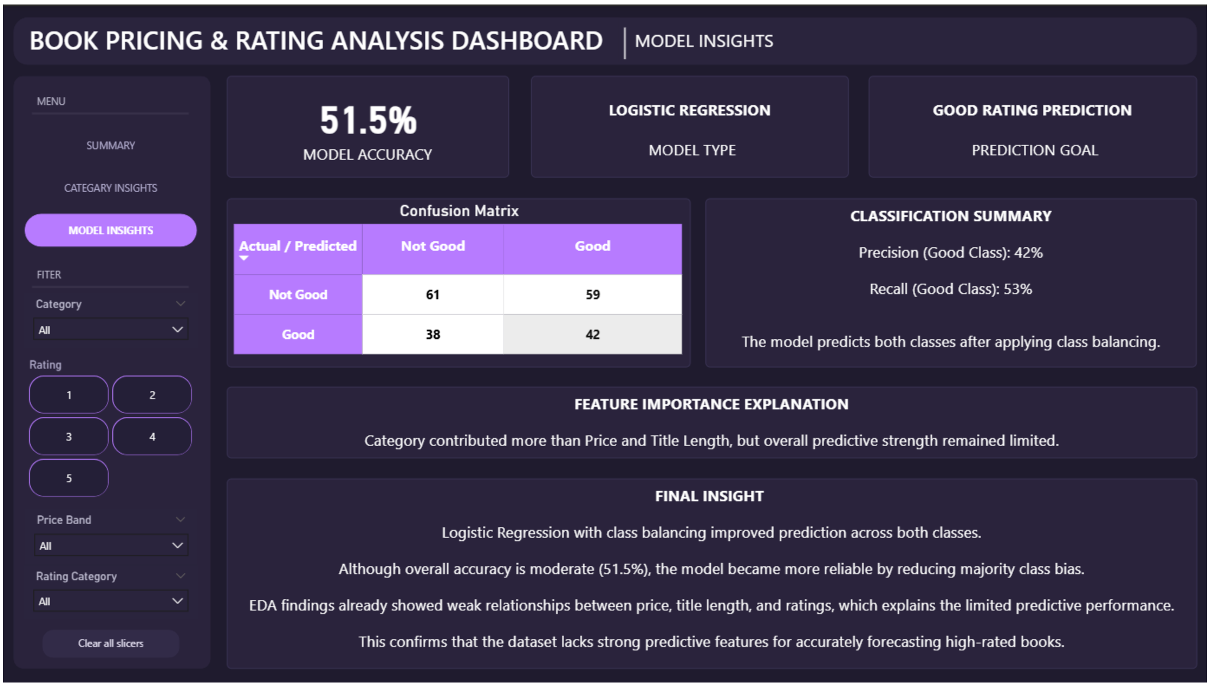
Page 2 proves that category selection matters more than pricing strategy.

Businesses should focus on strong-performing categories rather than assuming higher prices improve ratings.

PAGE 3: MODEL INSIGHTS DASHBOARD

Objective

The Model Insights Dashboard presents the machine learning model performance and explains whether book ratings can be predicted using available structured features.



Model Used

Logistic Regression

The project used Logistic Regression because:

- The target variable was binary (Good Rated or Not Good Rated)
- It is simple, interpretable, and suitable for classification
- It works well as a baseline classification model

This made it ideal for business-focused predictive analysis.

Prediction Goal

Good Rating Prediction

The model was built to predict whether a book would receive a Good Rating based on:

- Price
- Title Length
- Category

Model Accuracy

51.5%

The model achieved 51.5% accuracy after applying class balancing.

This indicates moderate prediction performance.

Confusion Matrix

Actual / Predicted	Not Good	Good
Not Good	61	59
Good	38	42

This shows that the model predicts both classes instead of only the majority class.

This is a better and more realistic model.

Classification Summary

Good Class Performance

- Precision → 42%
- Recall → 53%

This indicates that the model is able to identify good-rated books, but prediction strength remains limited.

Feature Importance Explanation

Category contributed more than Price and Title Length, but overall predictive strength remained weak.

This aligns with EDA findings where:

- Price had almost no correlation with Rating
- Title Length also showed weak influence

Final Insight

Logistic Regression with class balancing improved prediction across both classes, but overall model performance remained moderate because the dataset lacked strong predictive features such as:

- Reviews
- Author popularity

- Customer engagement
- Publication trends

This proves that book ratings cannot be accurately predicted using only price, title length, and category.

PHASE 6: KEY BUSINESS INSIGHTS

Major Findings

1. Price Does Not Drive Ratings

Higher-priced books are not necessarily better rated.

This means premium pricing does not guarantee customer satisfaction.

2. Category Has Stronger Influence

Certain categories consistently perform better than others.

Example:

Adult Fiction and Novels perform significantly better than Crime and Psychology

3. Most Books Are Average Rated

Only:

37.5% books were classified as Good Rated

This indicates quality concentration is limited.

4. Medium and High Price Bands Dominate

Most books are priced in Medium and High ranges, showing limited low-price competition.

5. Predicting Good Ratings Is Difficult

Without richer behavioral data, ratings remain difficult to forecast accurately.

PHASE 7: RECOMMENDATIONS

Business Recommendations

1. Focus More on High-Performing Categories

Publishers should invest more in strong-performing categories like:

- Adult Fiction
- Novels
- Christian Fiction

These categories show higher rating potential.

2. Improve Low-Performing Categories

Categories like:

- Crime
- Psychology
- Parenting

require quality improvement strategies.

3. Do Not Use Price as a Quality Indicator

Customers should not assume expensive books are better.

Businesses should focus on content quality over pricing strategy.

4. Collect More Behavioral Features

For better ML prediction, future datasets should include:

- Review count
- Customer feedback
- Author popularity
- Sales performance
- Publication trends

This would significantly improve predictive accuracy.

5. Use ML as Decision Support, Not Final Truth

The model should support decisions, not replace them, because predictive strength is currently limited.

FINAL CONCLUSION

This project successfully transformed raw web-scraped book data into meaningful business insights using data cleaning, EDA, machine learning, and an interactive Power BI dashboard.
It proved that category influences ratings more than price, while predicting high-rated books remains challenging due to limited strong predictive features.

Video Summary:
https://drive.google.com/file/d/1QfFr1BUY21ocvlQf3HprN56sU1vuVlGh/view?usp=sharing